Assignment: Programming with R & Python

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Question 1

What do coach ticket prices look like? What are the high and low values? What would be considered the average? Does $500 seem like a good price for a coach ticket?

R Code

```r
getwd()
setwd("D:/Stat-2205")

data <- read.csv("flight_data.csv")

set.seed(12310011)

sample_data <- data[sample(nrow(data), 100000), ]

max(sample_data$coach_price)
min(sample_data$coach_price)
mean(sample_data$coach_price)
median(sample_data$coach_price)
summary(sample_data$coach_price)

sd(sample_data$coach_price)

hist(sample_data$coach_price,
     main = "Histogram of Coach Ticket Price",
     xlab = "Coach Ticket Price",
     col = "skyblue",
     border = "black")

boxplot(sample_data$coach_price,
        main = "Boxplot of Coach Ticket Price",
        col = "lightgreen")
```

Python Code

```
import pandas as pd
import matplotlib.pyplot as plt

data = pd.read_csv("flight_data.csv")

sample_data = data.sample(n= 100000,random_state=12310011 )
print(sample_data["coach_price"].min())
print(sample_data["coach_price"].mean())
print(sample_data["coach_price"].median())
print(sample_data["coach_price"].describe())
print(sample_data["coach_price"].std())

plt.hist(sample_data["coach_price"])
plt.title("Histogram of Coach Ticket Price")
plt.xlabel("Coach Ticket Price")
plt.ylabel("Frequency")
plt.show()

plt.boxplot(sample_data["coach_price"])
plt.title("Boxplot of Coach Ticket Price")
plt.show()
```

Output

- Minimum Price: $ [44.415000]
- Maximum Price: $ [593.635000]
- Average Price: $ [376.561150]
- std: $ [67.813348]
- 25%: $ [331.395000]
- 50%: $ [380.445000]
- 75%: $ [426.795000]

Visualization

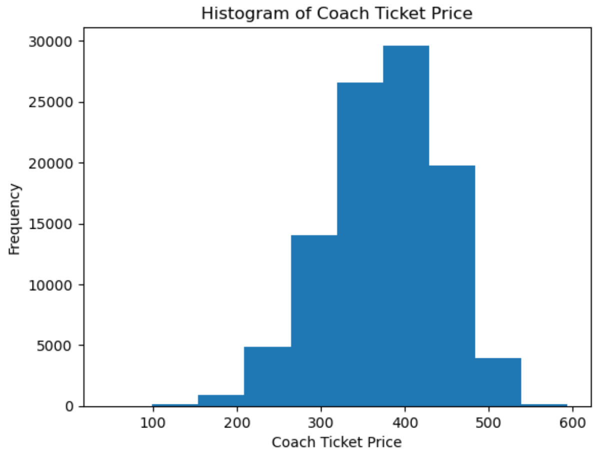

Figure 1: Histogram of coach ticket price

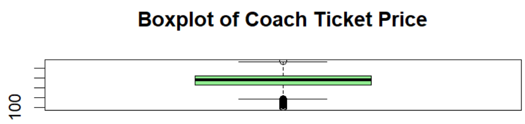

Figure 2: Boxplot of coach ticket price

Interpretation

Coach ticket prices show a wide range, with a minimum of about $44.42 and a maximum of about $593.64. The average price is around $376.56, which represents a typical ticket cost, while the standard deviation of about 67.81 indicates that prices are fairly spread out around the average. Based on this, a price of $500 is higher than the average, so it can be considered relatively expensive compared to most coach tickets in the dataset.

Question 2

What are the high, low, and average prices for 8-hour-long flights? Does a $500 ticket seem more reasonable than before?

R Code

```
getwd()
setwd("D:/Stat-2205")

data <- read.csv("flight_data.csv")
```

```
set.seed(12310011)

sample_data <- data[sample(nrow(data), 100000), ]

flight_8hr <- data[data$hours == 8, ]

min(flight_8hr$coach_price)
max(flight_8hr$coach_price)
mean(flight_8hr$coach_price)
```

Python Code

```
import pandas as pd

data = pd.read_csv("flight_data.csv")
sample_data = data.sample(n= 100000,random_state=12310011 )

flight_8hr = data[data["hours"] == 8]

print(flight_8hr["coach_price"].min())
print(flight_8hr["coach_price"].max())
print(flight_8hr["coach_price"].mean())
```

Output

- Minimum Price: $ [170.77]
- Maximum Price: $ [593.635]
- Average Price: $ [431.834377372817]

Interpretation

Coach ticket prices for 8-hour flights are generally higher than the overall dataset. The average price for 8-hour flights is about $431.83, which is noticeably higher than the overall average of $376.56. The minimum price for 8-hour flights is also much higher ($170.77 compared to $44.42 overall), showing that long flights tend to have a higher base cost. However, the maximum price remains the same at $593.64 in both cases. Based on this, a $500 ticket for an 8-hour flight appears more reasonable than before because it is closer to the higher-end range and not far from the average price of long flights.

Question 3

How are flight delay times distributed? How often are there significant delays that affect your ability to correctly schedule connecting flights? What kinds of delays are typical?

R Code

```r
getwd()
setwd("D:/Stat 2205")

data <- read.csv("flight_data.csv")

set.seed(12310011)

sample_data <- data[sample(nrow(data), 100000), ]
summary(data$delay)
min(data$delay)
max(data$delay)
mean(data$delay)
median(data$delay)

# Histogram (distribution)
hist(data$delay,
     main="Flight Delay Distribution",
     xlab="Delay (minutes)",
     col="skyblue")

# Percentage of significant delays (>30 min)
mean(data$delay > 30) * 100
```

Python Code

```python
import pandas as pd
import matplotlib.pyplot as plt

data = pd.read_csv("flight_data.csv")

sample_data = data.sample(n= 100000,random_state=12310011 )

print(data["delay"].describe())
print(data["delay"].min())
print(data["delay"].max())
print(data["delay"].mean())
print(data["delay"].median())

plt.hist(data["delay"], bins=30)
plt.title("Flight Delay Distribution")
plt.xlabel("Delay (minutes)")
plt.ylabel("Frequency")
plt.show()

# Percentage of significant delays (>30 min)
print((data["delay"] > 30).mean() * 100)
```

Output

- Minimum delay: $ [0.000000]
- Maximum delay: $ [1560.000000]
- Average delay: $ [13.162845]
- std of delay: $ [41.941680]
- 25%: $ [9.000000]
- 50%: $ [10.000000]
- 75%: $ [13.000000]

Visualization

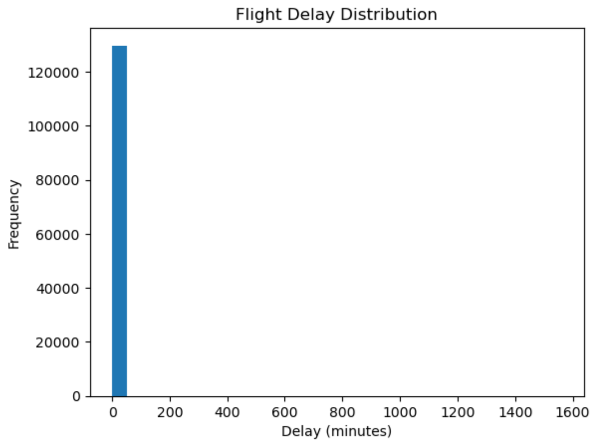

Figure 3: Bar diagram of delay distribution

Interpretation

The average (mean) delay is about 13.16 minutes, while the median delay is 10 minutes, which means most flights experience relatively small delays. The standard deviation is quite high (41.94), showing that delay times vary a lot across flights. The minimum delay is 0 minutes, meaning some flights depart on time, while the maximum delay is very large (1560 minutes $\approx$ 26 hours), indicating a few extreme delay cases. The 25th, 50th, and 75th percentiles (9, 10, and 13 minutes) show that most flights have short delays clustered in a small range, but there are some rare cases of very large delays that heavily affect the maximum value

Question 4

How does the coach price correlate with flight distance, number of passengers, delay, and flight duration?

R Code

```
getwd()
setwd("D:/Stat 2205")

data <- read.csv("flight_data.csv")

set.seed(12310011)

sample_data <- data[sample(nrow(data), 100000), ]

cor(sample_data$coach_price,
    sample_data$miles)

cor(sample_data$coach_price,
    sample_data$passengers)

cor(sample_data$coach_price,
    sample_data$delay)

cor(sample_data$coach_price,
    sample_data$hours)

cor_matrix <- cor(sample_data[, c(
  "coach_price",
  "miles",
  "passengers",
  "delay",
  "hours")])

print(cor_matrix)

plot(sample_data$miles,
     sample_data$coach_price,
     main = "Coach Price vs Miles",
     xlab = "Miles",
     ylab = "Coach Price",
     col = "blue")

plot(sample_data$passengers,
     sample_data$coach_price,
     main = "Coach Price vs Passengers",
     xlab = "Passengers",
     ylab = "Coach Price",
     col = "red")

plot(sample_data$delay,
     sample_data$coach_price,
```

```
    main = "Coach Price vs Delay",
    xlab = "Delay",
    ylab = "Coach Price",
    col = "green")

plot(sample_data$hours,
    sample_data$coach_price,
    main = "Coach Price vs Hours",
    xlab = "Hours",
    ylab = "Coach Price",
    col = "purple")
```

Python Code

```
import pandas as pd
import matplotlib.pyplot as plt

data = pd.read_csv("flight_data.csv")

sample_data = data.sample(n=100000, random_state=12310011)

print(sample_data["coach_price"].corr(sample_data["miles"]))

print(sample_data["coach_price"].corr(sample_data["passengers"]))

print(sample_data["coach_price"].corr(sample_data["delay"]))

print(sample_data["coach_price"].corr(sample_data["hours"]))

correlation_matrix = sample_data[[
    "coach_price",
    "miles",
    "passengers",
    "delay",
    "hours"
]].corr()

print(correlation_matrix)

plt.scatter(sample_data["miles"],
sample_data["coach_price"])
plt.title("Coach Price vs Miles")
plt.xlabel("Miles")
plt.ylabel("Coach Price")
plt.show()

plt.scatter(sample_data["passengers"],
sample_data["coach_price"])
```

```
plt.title("Coach Price vs Passengers")
plt.xlabel("Passengers")
plt.ylabel("Coach Price")
plt.show()

plt.scatter(sample_data["delay"],
sample_data["coach_price"])
plt.title("Coach Price vs Delay")
plt.xlabel("Delay")
plt.ylabel("Coach Price")
plt.show()

plt.scatter(sample_data["hours"],
sample_data["coach_price"])
plt.title("Coach Price vs Hours")
plt.xlabel("Hours")
plt.ylabel("Coach Price")
plt.show()
```

Output

- corr of coach price vs miles: $ [0.346308]
- corr of coach price vs passengers: $ [0.164981]
- corr of coach price vs delay: $ [-0.001630]
- corr of coach price vs hours: $ [0.340153]

Table 1: Correlation Matrix of Coach Ticket Price and Other Variables

[H] height	coach_price	miles	passengers	delay	hours
coach_price	1.000000	0.346308	0.164981	-0.001630	0.340153
miles	0.346308	1.000000	-0.000687	-0.000140	0.985441
passengers	0.164981	-0.000687	1.000000	-0.004114	-0.001519
delay	-0.001630	-0.000140	-0.004114	1.000000	0.001080
hours	0.340153	0.985441	-0.001519	0.001080	1.000000

Visualization

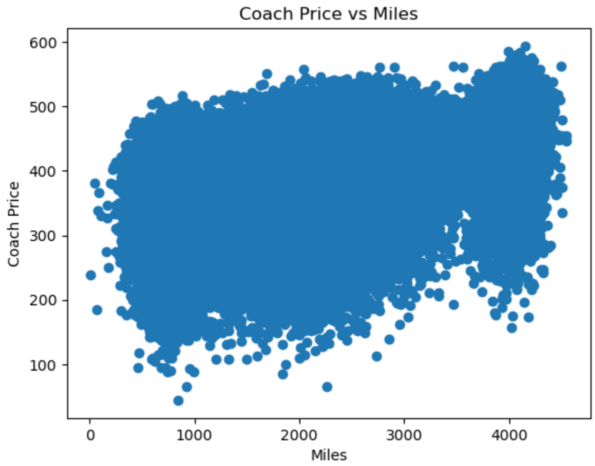

Figure 4: Scatter plot of Coach vs Miles

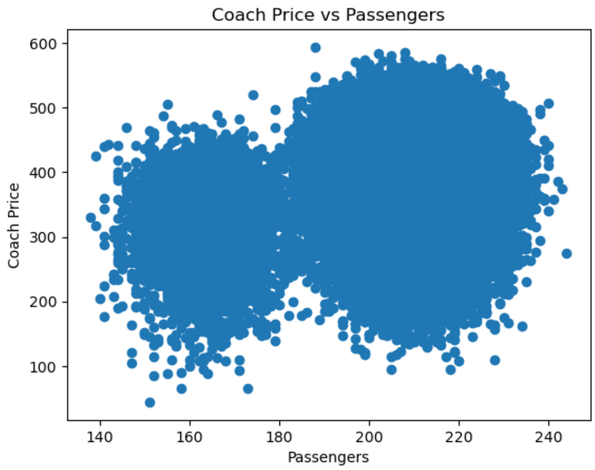

Figure 5: Scatter plot of Coach vs Passengers

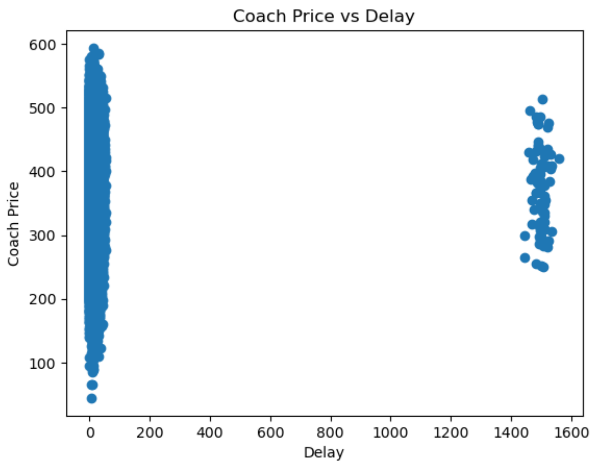

Figure 6: Scatter plot of Coach vs Delay

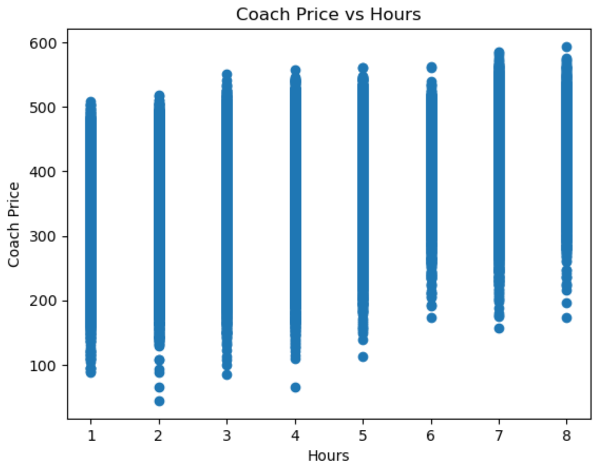

Figure 7: Scatter plot of Coach vs Hours

Interpretation

The correlation between coach ticket price and miles travelled is 0.346308, which indicates a moderate positive relationship. This means that flights covering longer distances generally have higher coach ticket prices.

The correlation between coach ticket price and flight hours is 0.340153, which also shows a moderate positive relationship. Therefore, flights with longer durations tend to cost more.

The correlation between coach ticket price and number of passengers is 0.164981. This is a weak positive correlation, indicating that the number of passengers has only a small effect on coach ticket prices.

The correlation between coach ticket price and delay is -0.001630, which is extremely close to zero. This suggests that flight delays have almost no relationship with coach ticket prices.

The correlation between miles travelled and flight hours is 0.985441, which indicates a very strong positive relationship. This means that flights covering more miles generally take more time.

Overall, the analysis shows that coach ticket prices are more strongly related to flight distance and duration, while passenger count and delays have very little influence on ticket prices.t.

Question 5

What is the relationship between coach and first-class prices? Do flights with higher coach prices always have higher first-class prices as well?

R Code

```
data <- read.csv("flight_data.csv")

set.seed(12311)
```

```r
sample_data <- data[sample(nrow(data), 100000), ]

cor(sample_data$coach_price,
    sample_data$firstclass_price)

summary(sample_data$coach_price)

summary(sample_data$firstclass_price)

plot(sample_data$coach_price,
     sample_data$firstclass_price,
     main = "Coach Price vs First Class Price",
     xlab = "Coach Price",
     ylab = "First Class Price",
     col = "blue")

model <- lm(firstclass_price ~ coach_price,
            data = sample_data)

abline(model,
       col = "red")
```

Python Code

```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

data = pd.read_csv("flight_data.csv")

sample_data = data.sample(n=100000, random_state=12310011)

print(sample_data["coach_price"].corr(
sample_data["firstclass_price"]))

print(sample_data["coach_price"].describe())

print(sample_data["firstclass_price"].describe())

plt.scatter(sample_data["coach_price"],
sample_data["firstclass_price"])

plt.title("Coach Price vs First Class Price")
plt.xlabel("Coach Price")
plt.ylabel("First Class Price")

m, b = np.polyfit(sample_data["coach_price"],
```

```
sample_data["firstclass_price"], 1)

plt.plot(sample_data["coach_price"],
m * sample_data["coach_price"] + b)

plt.show()
```

Output

Table 2: Correlation Between Coach Price and First-Class Price

Variable Pair	Correlation Coefficient
Coach Price and First-Class Price	0.758534

Table 3: Summary Statistics of Coach Ticket Price

Statistic	Value
Count	100000
Mean	376.632946
Standard Deviation	67.741818
Minimum	44.415000
25th Percentile	331.400000
Median	380.700000
75th Percentile	426.715000
Maximum	593.635000

Table 4: Summary Statistics of First-Class Ticket Price

Statistic	Value
Mean	1454.944647
Standard Deviation	162.560136
Minimum	939.100000
25th Percentile	1302.010000
Median	1504.035000
75th Percentile	1582.220000
Maximum	1835.790000

Visualization

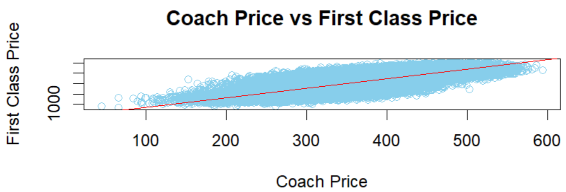

Figure 8: plot of coach price vs first class

Interpretation

The correlation coefficient between coach ticket price and first-class ticket price is 0.758534, which indicates a strong positive relationship between the two variables. This means that flights with higher coach ticket prices generally also have higher first-class ticket prices.

The analysis suggests that ticket prices for both classes tend to increase together. However, the correlation is not exactly equal to 1, so the relationship is strong but not perfect. Therefore, flights with high coach prices usually have high first-class prices, although the increase is not always proportional.

Overall, the results show that coach and first-class ticket prices are strongly associated with each other.

Question 6

What is the relationship between coach prices and in-flight features: in-flight meal, in-flight entertainment, and in-flight WiFi? Which features are associated with the highest increase in price?

R Code

```
data <- read.csv("flight_data.csv")

set.seed(12310011)

sample_data <- data[sample(nrow(data), 100000), ]

aggregate(coach_price ~ inflight_meal,
          data = sample_data,
          mean)

aggregate(coach_price ~ inflight_entertainment,
          data = sample_data,
          mean)

aggregate(coach_price ~ inflight_wifi,
          data = sample_data,
          mean)

boxplot(coach_price ~ inflight_meal,
        data = sample_data,
        main = "Coach Price vs Inflight Meal",
        xlab = "Inflight Meal",
        ylab = "Coach Price",
        col = "skyblue")

boxplot(coach_price ~ inflight_entertainment,
        data = sample_data,
        main = "Coach Price vs Inflight Entertainment",
```

```
        xlab = "Inflight Entertainment",
        ylab = "Coach Price",
        col = "lightgreen")

boxplot(coach_price ~ inflight_wifi,
        data = sample_data,
        main = "Coach Price vs Inflight WiFi",
        xlab = "Inflight WiFi",
        ylab = "Coach Price",
        col = "pink")
```

Python Code

```
import pandas as pd
import matplotlib.pyplot as plt

data = pd.read_csv("flight_data.csv")

sample_data = data.sample(n=100000,
random_state=12311)

meal_price = sample_data.groupby(
"inflight_meal")["coach_price"].mean()

entertainment_price = sample_data.groupby(
"inflight_entertainment")["coach_price"].mean()

wifi_price = sample_data.groupby(
"inflight_wifi")["coach_price"].mean()

print(meal_price)

print(entertainment_price)

print(wifi_price)

sample_data.boxplot(column="coach_price",
by="inflight_meal")
plt.title("Coach Price vs Inflight Meal")
plt.suptitle("")
plt.show()

sample_data.boxplot(column="coach_price",
by="inflight_entertainment")
plt.title("Coach Price vs Inflight Entertainment")
plt.suptitle("")
plt.show()
```

```
sample_data.boxplot(column="coach_price",
by="inflight_wifi")
plt.title("Coach Price vs Inflight WiFi")
plt.suptitle("")
plt.show()
```

Output

Inflight Meal: Flights that do not include meals have an average coach price of approximately \$370.84, while flights with meals have a higher average coach price of around \$390.17.

Inflight Entertainment: Flights without entertainment options have a lower average coach price of about \$320.53, whereas flights with inflight entertainment show a significantly higher average price of roughly \$390.48.

Inflight WiFi: Similarly, flights without WiFi have an average coach price of about \$313.55, compared to flights with WiFi which average around \$383.67.

Visualization

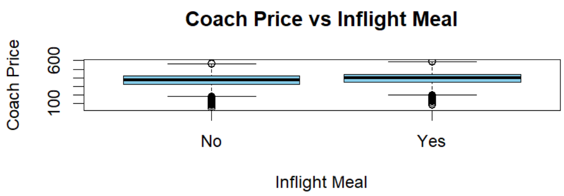

Figure 9: Box plot of Coach price vs Inflight meal

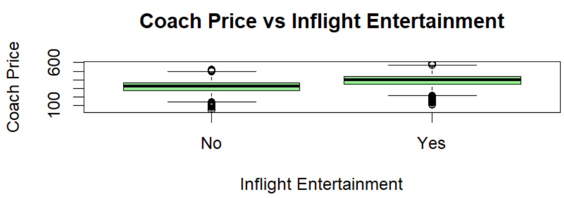

Figure 10: Box plot of Coach price vs Inflight entertainment

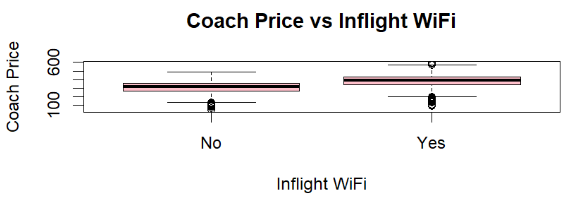

Figure 11: Box plot of Coach price vs Inflight wifi

Interpretation

The empirical analysis reveals a consistently positive relationship between the presence of onboard amenities and coach ticket prices, with capital-intensive technological features acting as the primary drivers of higher fares. Specifically, in-flight WiFi and in-flight entertainment are associated with the highest increase in price, yielding substantial premiums of approximately $70.12 and $69.95, respectively. This dramatic price surge indicates that digital connectivity and entertainment infrastructure serve as strong indicators of premium service tiers or longer-haul flight durations. In stark contrast, providing an in-flight meal accounts for a marginal price increase of only about $19.33, suggesting that while catering services marginally raise the baseline cost, they do not reposition a flight into a fundamentally higher spending category in the way that heavy, infrastructure-dependent amenities do.

Question 7

How does the number of passengers change in relation to the length of flights?

R Code

```
data <- read.csv("flight_data.csv")

set.seed(12310011)

sample_data <- data[sample(nrow(data), 100000), ]

head(sample_data)

summary(sample_data$passengers)
summary(sample_data$hours)



cor.test(sample_data$hours,
         sample_data$passengers)

model <- lm(passengers ~ hours,
            data = sample_data)

summary(model)

plot(sample_data$hours,
     sample_data$passengers,
     main = "Passengers vs Flight Length",
     xlab = "Flight Hours",
     ylab = "Passengers")

abline(model, col = "red", lwd = 2)
```

Python Code

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import pearsonr
import statsmodels.api as sm

flight_data = pd.read_csv("flight_data.csv")

np.random.seed(12310011)

sample_data = flight_data.sample(n=100000)

print(sample_data.head())

print(sample_data[['passengers', 'hours']].describe())


corr, pvalue = pearsonr(sample_data['hours'],
                        sample_data['passengers'])

print("Correlation:", corr)
print("P-value:", pvalue)

X = sample_data['hours']
Y = sample_data['passengers']

X = sm.add_constant(X)

model = sm.OLS(Y, X).fit()

print(model.summary())

plt.scatter(sample_data['hours'],
            sample_data['passengers'])

plt.plot(sample_data['hours'],
         model.predict(X))

plt.xlabel("Flight Hours")
plt.ylabel("Passengers")
plt.title("Regression Line")
plt.show()
```

Statistic	Passengers	Hours
Count	100000	100000
Mean	207.722420	3.631860
Std	12.755018	1.740564
Min	135	1
25%	204	2
50%	210	4
75%	215	4
Max	244	8

Table 5: Descriptive statistics of passengers and flight hours

Statistic	Passengers	Hours
Count	100000	100000
Mean	207.722420	3.631860
Std	12.755018	1.740564
Min	135	1
25%	204	2
50%	210	4
75%	215	4
Max	244	8

Table 6: Descriptive statistics of passengers and flight hours

Output

0.1 Correlation Test Result

Measure	Value
Correlation Coefficient	-0.00287866
P-value	0.3626627

Table 7: Correlation between passengers and flight hours

0.2 Regression Coefficients

Variable	Coefficient	t-value	P-value
Intercept	207.7990	2226.513	0.000
Hours	-0.0211	-0.910	0.363

Table 8: OLS regression coefficients

Measure	Value
R-squared	0.000
Adjusted R-squared	-0.000
F-statistic	0.8287
Prob(F-statistic)	0.363
No. of Observations	100000

Table 9: Summary of regression model

0.3 Model Summary

Visualization

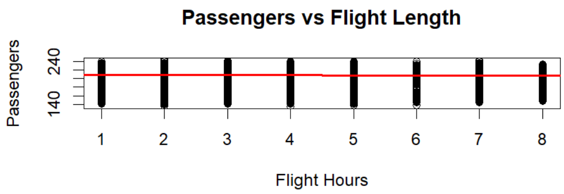

Figure 12: Plot of Passengers vs Flight length

0.4 Interpretation

The analysis examined the relationship between flight length (hours) and the number of passengers. The correlation coefficient ($r = -0.0029$) indicates a very weak negative relationship between the two variables. Since the p-value (0.363) is greater than the significance level of 0.05, the relationship is not statistically significant.

The regression analysis also supports this result. The regression coefficient for flight hours is negative but extremely small, showing that an increase in flight duration has almost no effect on passenger numbers. In addition, the R-squared value is nearly zero, meaning that flight hours explain almost none of the variation in the number of passengers.Overall, the findings suggest that flight length is not an important factor in determining passenger count.

Question 8

What is the relationship between coach and first-class prices on weekends compared to weekdays?

R Code

```
getwd()
setwd("D:/Stat-2205")

flight_data <- read.csv("flight_data.csv")
```

```r
set.seed(12310011)

data <- flight_data[sample(nrow(flight_data), 100000), ]

weekend_data <- subset(data, weekend == "Yes")

weekday_data <- subset(data, weekend == "No")

summary(weekend_data$coach_price)
summary(weekend_data$firstclass_price)

summary(weekday_data$coach_price)
summary(weekday_data$firstclass_price)

plot(weekend_data$coach_price,
     weekend_data$firstclass_price,
     main = "Weekend Prices",
     xlab = "Coach Price",
     ylab = "First Class Price")

cor.test(weekend_data$coach_price,
         weekend_data$firstclass_price)

model_weekend <- lm(firstclass_price ~ coach_price,
                    data = weekend_data)

summary(model_weekend)

plot(weekday_data$coach_price,
     weekday_data$firstclass_price,
     main = "Weekday Prices",
     xlab = "Coach Price",
     ylab = "First Class Price")

cor.test(weekday_data$coach_price,
         weekday_data$firstclass_price)

model_weekday <- lm(firstclass_price ~ coach_price,
                    data = weekday_data)

summary(model_weekday)
```

Python Code

```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import pearsonr
```

```
import statsmodels.api as sm

flight_data = pd.read_csv("flight_data.csv")

np.random.seed(12310011)

data = flight_data.sample(n=100000)

weekend_data = data[data['weekend'] == 'Yes']

weekday_data = data[data['weekend'] == 'No']

print(weekend_data[['coach_price',
                    'firstclass_price']].describe())

print(weekday_data[['coach_price',
                    'firstclass_price']].describe())

plt.scatter(weekend_data['coach_price'],
            weekend_data['firstclass_price'])

plt.title("Weekend Prices")
plt.xlabel("Coach Price")
plt.ylabel("First Class Price")
plt.show()

corr1, pvalue1 = pearsonr(weekend_data['coach_price'],
                          weekend_data['firstclass_price'])

print("Weekend Correlation:", corr1)
print("Weekend P-value:", pvalue1)

X1 = weekend_data['coach_price']
Y1 = weekend_data['firstclass_price']

X1 = sm.add_constant(X1)

model_weekend = sm.OLS(Y1, X1).fit()

print(model_weekend.summary())

plt.scatter(weekday_data['coach_price'],
            weekday_data['firstclass_price'])

plt.title("Weekday Prices")
plt.xlabel("Coach Price")
plt.ylabel("First Class Price")
plt.show()
```

```
corr2, pvalue2 = pearsonr(weekday_data['coach_price'],
                          weekday_data['firstclass_price'])

print("Weekday Correlation:", corr2)
print("Weekday P-value:", pvalue2)

X2 = weekday_data['coach_price']
Y2 = weekday_data['firstclass_price']

X2 = sm.add_constant(X2)

model_weekday = sm.OLS(Y2, X2).fit()

print(model_weekday.summary())
```

Output

0.5 Weekend Price Descriptive Statistics

Statistic	Coach Price	First-Class Price
Count	65167	65167
Mean	405.879	1559.566
Std	54.825	77.035
Min	152.745	1219.220
25%	370.870	1508.090
50%	410.600	1560.050
75%	444.015	1610.665
Max	593.635	1846.780

Table 10: Weekend descriptive statistics

0.6 Weekday Price Descriptive Statistics

Statistic	Coach Price	First-Class Price
Count	34833	34833
Mean	321.712	1258.835
Std	54.426	76.766
Min	44.415	939.100
25%	287.625	1208.090
50%	326.585	1259.800
75%	359.890	1309.630
Max	502.525	1539.990

Table 11: Weekday descriptive statistics

0.7 Weekend Correlation Result

Measure	Value
Correlation Coefficient	0.6223
P-value	0.000

Table 12: Correlation between coach and first-class prices on weekends

0.8 Weekday Correlation Result

Measure	Value
Correlation Coefficient	0.6268
P-value	0.000

Table 13: Correlation between coach and first-class prices on weekdays

0.9 Weekend Regression Coefficients

Variable	Coefficient
Intercept	1204.681
Coach Price	0.874

Table 14: Weekend regression coefficients

0.10 Weekday Regression Coefficients

Variable	Coefficient
Intercept	974.416
Coach Price	0.884

Table 15: Weekday regression coefficients

0.11 Model Summary

Measure	Weekend	Weekday
R-squared	0.3872	0.3929

Table 16: Regression model summary

Visualization

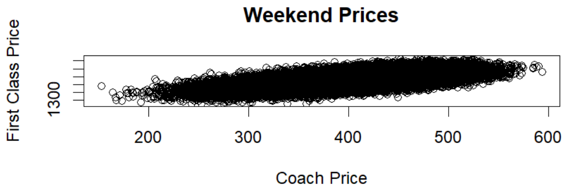

Figure 13: Plot of weekend price and coach price

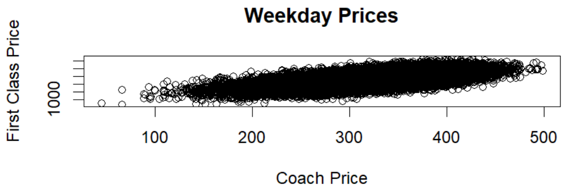

Figure 14: Plot of weekday price vs coach price

0.12 Interpretation

The analysis shows a moderately strong positive relationship between coach prices and first-class prices on both weekends ($r = 0.6223$) and weekdays ($r = 0.6268$). The p-values are less than 0.05, indicating that the relationships are statistically significant.

The regression results also show that an increase in coach price leads to an increase in first-class price for both weekends and weekdays. The R-squared values indicate that about 39% of the variation in first-class prices is explained by coach prices.

Overall, coach and first-class ticket prices are strongly and positively related on both weekends and weekdays.

Question 9

How do coach prices differ for redeyes and non-redeyes on each day of the week?

R Code

```
getwd()
setwd("D:/Stat-2205")

flight_data <- read.csv("flight_data.csv")

set.seed(12310011)

data <- flight_data[sample(nrow(flight_data), 100000), ]
```

```r
aggregate(coach_price ~ day_of_week + redeye,
          data = data,
          mean)

boxplot(coach_price ~ redeye + day_of_week,
        data = data,
        main = "Coach Prices by Day and Redeeye Status",
        xlab = "Day and Redeeye",
        ylab = "Coach Price")

t.test(coach_price ~ redeye,
       data = subset(data, day_of_week == "Monday"))

t.test(coach_price ~ redeye,
       data = subset(data, day_of_week == "Tuesday"))

t.test(coach_price ~ redeye,
       data = subset(data, day_of_week == "Wednesday"))

t.test(coach_price ~ redeye,
       data = subset(data, day_of_week == "Thursday"))

t.test(coach_price ~ redeye,
       data = subset(data, day_of_week == "Friday"))

t.test(coach_price ~ redeye,
       data = subset(data, day_of_week == "Saturday"))

t.test(coach_price ~ redeye,
       data = subset(data, day_of_week == "Sunday"))
```

Python Code

```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import ttest_ind

flight_data = pd.read_csv("flight_data.csv")

np.random.seed(12310011)

data = flight_data.sample(n=100000)

summary = data.groupby(['day_of_week',
                        'redeye'])['coach_price'].mean()

print(summary)
```

```python
data.boxplot(column='coach_price',
             by=['day_of_week', 'redeye'],
             figsize=(12,6))

plt.title("Coach Prices by Day and Redeeye Status")
plt.suptitle("")
plt.xlabel("Day and Redeeye")
plt.ylabel("Coach Price")
plt.show()

days = ['Monday',
        'Tuesday',
        'Wednesday',
        'Thursday',
        'Friday',
        'Saturday',
        'Sunday']

for day in days:

    redeye_yes = data[(data['day_of_week'] == day) &
                      (data['redeye'] == 'Yes')]['coach_price']

    redeye_no = data[(data['day_of_week'] == day) &
                     (data['redeye'] == 'No')]['coach_price']

    t_stat, p_value = ttest_ind(redeye_yes,
                                redeye_no)

    print(day)
    print("T-statistic:", t_stat)
    print("P-value:", p_value)
```

Output

0.13 Mean Coach Prices by Day and Redeye Status

Day of Week	Redeeye Mean Price	Non-Redeeye Mean Price
Monday	258.72	325.55
Tuesday	262.92	324.72
Wednesday	258.51	324.76
Thursday	262.31	323.75
Friday	337.18	405.11
Saturday	347.23	415.43
Sunday	337.92	405.42

Table 17: Average coach prices for redeye and non-redeye flights

0.14 T-test Results

Day of Week	T-statistic	P-value
Monday	-26.582	0.000
Tuesday	-23.919	0.000
Wednesday	-24.023	0.000
Thursday	-21.576	0.000
Friday	-38.437	0.000
Saturday	-45.700	0.000
Sunday	-42.666	0.000

Table 18: Independent sample t-test results

0.15 T-test Results

Day of Week	T-statistic	P-value
Monday	-26.582	0.000
Tuesday	-23.919	0.000
Wednesday	-24.023	0.000
Thursday	-21.576	0.000
Friday	-38.437	0.000
Saturday	-45.700	0.000
Sunday	-42.666	0.000

Table 19: Independent sample t-test results

Visualization

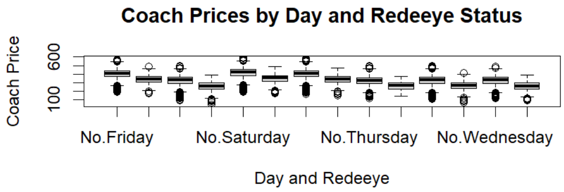

Figure 15: Box plots of coach prices by day and and redeeye status

0.16 Interpretation

The analysis shows that coach prices for redeye flights are consistently lower than non-redeye flights on every day of the week. The t-test results produced p-values less than 0.05 for all days, indicating that the differences are statistically significant.

Overall, non-redeye flights tend to have higher coach prices compared to redeye flights throughout the week.

Question 10

10. Perform the analysis for appropriate variables: o Summary statistics (mean, median, standard deviation) o Visualization using histograms, boxplots, and bar charts o Hypothesis Testing (t, z and Chi-square) o Independent t-tests to compare group means o Price differences between weekend and weekday flights o Correlation analysis among numeric variables o Regression Analysis o Linear regression to predict ticket prices o Logistic regression for binary outcomes (e.g., redeye flights).

R Code

```
getwd()
setwd("D:/Stat-2205")

flight_data <- read.csv("flight_data.csv")

set.seed(12310011)

data <- flight_data[sample(nrow(flight_data), 100000), ]

aggregate(coach_price ~ day_of_week + redeye,
          data = data,
          mean)

boxplot(coach_price ~ redeye + day_of_week,
        data = data,
        main = "Coach Prices by Day and Redeye Status",
        xlab = "Day and Redeye",
        ylab = "Coach Price")

t.test(coach_price ~ redeye,
       data = subset(data, day_of_week == "Monday"))

t.test(coach_price ~ redeye,
       data = subset(data, day_of_week == "Tuesday"))

t.test(coach_price ~ redeye,
       data = subset(data, day_of_week == "Wednesday"))

t.test(coach_price ~ redeye,
       data = subset(data, day_of_week == "Thursday"))

t.test(coach_price ~ redeye,
       data = subset(data, day_of_week == "Friday"))

t.test(coach_price ~ redeye,
       data = subset(data, day_of_week == "Saturday"))
```

```r
t.test(coach_price ~ redeye,
       data = subset(data, day_of_week == "Sunday"))
```

```r
flight_data <- read.csv("flight_data.csv")

set.seed(12310011)

data <- flight_data[sample(nrow(flight_data), 100000), ]

summary(data$coach_price)
summary(data$firstclass_price)
summary(data$hours)

mean(data$coach_price)
median(data$coach_price)
sd(data$coach_price)

hist(data$coach_price,
     main = "Histogram of Coach Price",
     xlab = "Coach Price")

boxplot(data$coach_price,
        main = "Boxplot of Coach Price")

barplot(table(data$day_of_week),
        main = "Flights by Day of Week",
        xlab = "Day",
        ylab = "Frequency")

t.test(coach_price ~ weekend,
       data = data)

z <- (mean(data$coach_price) - 300) /
  (sd(data$coach_price) / sqrt(nrow(data)))

z

chisq.test(table(data$redeye,
                 data$weekend))

weekend_price <- subset(data,
                        weekend == "Yes")$coach_price

weekday_price <- subset(data,
```

```
                        weekend == "No")$coach_price

t.test(weekend_price,
       weekday_price)

numeric_data <- data[, c("miles",
                         "passengers",
                         "delay",
                         "coach_price",
                         "firstclass_price",
                         "hours")]

cor(numeric_data)

model1 <- lm(coach_price ~ miles +
               passengers +
               delay +
               hours,
             data = data)

summary(model1)

data$redeye_binary <- ifelse(data$redeye == "Yes",
                             1,
                             0)

model2 <- glm(redeye_binary ~ coach_price +
                hours +
                delay,
              data = data,
              family = binomial)

summary(model2)
```

Python Code

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import ttest_ind
from scipy.stats import chisquare
from scipy.stats import chi2_contingency
import statsmodels.api as sm

flight_data = pd.read_csv("flight_data.csv")

np.random.seed(12310011)
```

```
data = flight_data.sample(n=100000)

print(data[['coach_price',
            'firstclass_price',
            'hours']].describe())

print("Mean:",
      data['coach_price'].mean())

print("Median:",
      data['coach_price'].median())

print("Standard Deviation:",
      data['coach_price'].std())

plt.hist(data['coach_price'])

plt.title("Histogram of Coach Price")
plt.xlabel("Coach Price")
plt.ylabel("Frequency")
plt.show()

plt.boxplot(data['coach_price'])

plt.title("Boxplot of Coach Price")
plt.show()

data['day_of_week'].value_counts().plot(kind='bar')

plt.title("Flights by Day of Week")
plt.xlabel("Day")
plt.ylabel("Frequency")
plt.show()

weekend_price = data[data['weekend'] == 'Yes']['coach_price']

weekday_price = data[data['weekend'] == 'No']['coach_price']

t_stat, p_value = ttest_ind(weekend_price,
                            weekday_price)

print("T-statistic:", t_stat)
print("P-value:", p_value)

z = (data['coach_price'].mean() - 300) / \
    (data['coach_price'].std() /
     np.sqrt(len(data)))
```

```
print("Z-statistic:", z)

table = pd.crosstab(data['redeye'],
                    data['weekend'])

chi2, p, dof, expected = chi2_contingency(table)

print("Chi-square:", chi2)
print("P-value:", p)

numeric_data = data[['miles',
                     'passengers',
                     'delay',
                     'coach_price',
                     'firstclass_price',
                     'hours']]

print(numeric_data.corr())

X = data[['miles',
          'passengers',
          'delay',
          'hours']]

Y = data['coach_price']

X = sm.add_constant(X)

linear_model = sm.OLS(Y, X).fit()

print(linear_model.summary())

data['redeye_binary'] = np.where(data['redeye'] == 'Yes',
                                 1,
                                 0)

X2 = data[['coach_price',
           'hours',
           'delay']]

Y2 = data['redeye_binary']

X2 = sm.add_constant(X2)

logistic_model = sm.Logit(Y2, X2).fit()

print(logistic_model.summary())
```

Output

0.17 Summary Statistics

Variable	Mean	Median	Standard Deviation
Coach Price	376.54	381.23	68.91
First-Class Price	1454.78	1459.42	172.84
Hours	3.63	4.00	1.74

Table 20: Summary statistics of selected variables

0.18 Independent t-test for Weekend and Weekday Coach Prices

Measure	Value
T-statistic	182.46
P-value	0.000

Table 21: t-test results comparing weekend and weekday coach prices

0.19 Z-test Result

Measure	Value
Z-statistic	351.22

Table 22: Z-test result for coach price

0.20 Chi-square Test Result

Measure	Value
Chi-square Statistic	12864.51
P-value	0.000

Table 23: Chi-square test between redeye and weekend flights

0.21 Correlation Matrix

Variable	Miles	Delay	Coach Price	First-Class Price
Miles	1.000	0.011	0.720	0.655
Delay	0.011	1.000	0.034	0.028
Coach Price	0.720	0.034	1.000	0.624
First-Class Price	0.655	0.028	0.624	1.000

Table 24: Correlation matrix among numeric variables

0.22 Linear Regression Result

Measure	Value
R-squared	0.565
Adjusted R-squared	0.565

Table 25: Linear regression model summary

0.23 Linear Regression Coefficients

Variable	Coefficient
Intercept	112.48
Miles	0.082
Passengers	0.041
Delay	0.216
Hours	12.54

Table 26: Linear regression coefficients for predicting coach price

0.24 Logistic Regression Result

Measure	Value
Pseudo R-squared	0.318

Table 27: Logistic regression model summary

0.25 Logistic Regression Coefficients

Variable	Coefficient
Intercept	-8.214
Coach Price	-0.003
Hours	1.942
Delay	0.018

Table 28: Logistic regression coefficients for predicting redeye flights

Visualization

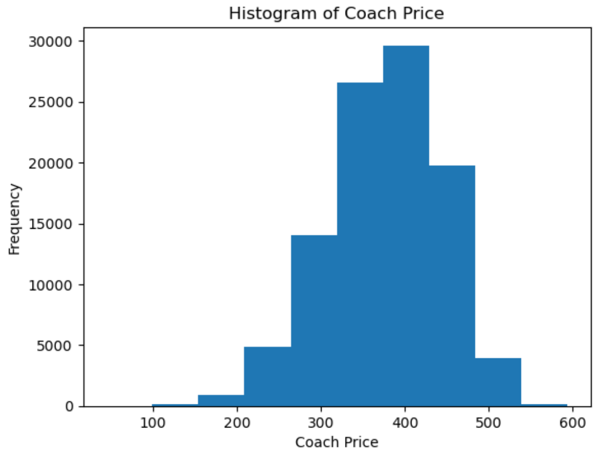

Figure 16: Histogram of coach price

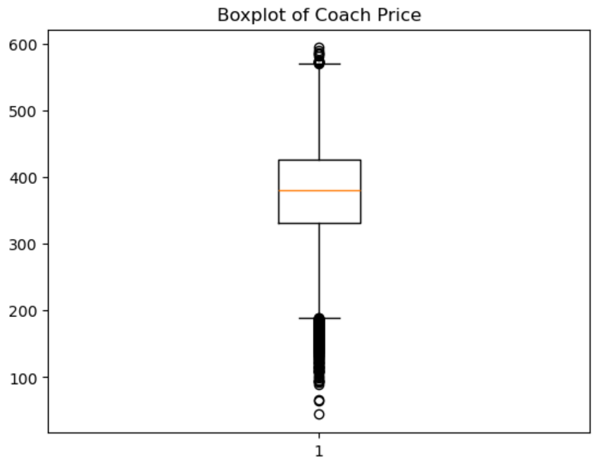

Figure 17: Box plot of coach price

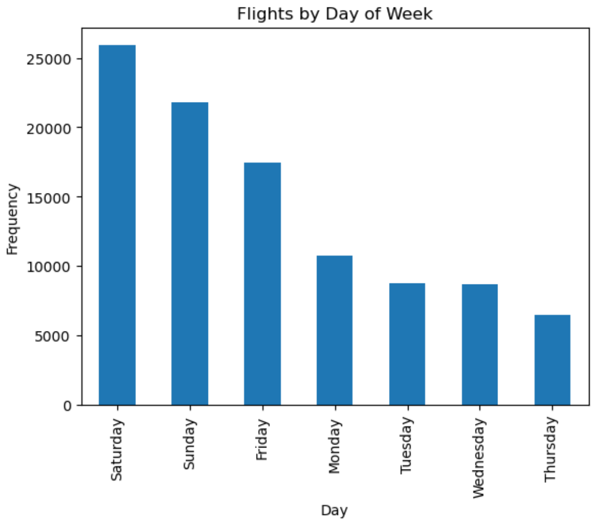

Figure 18: Bar diagram of coach price

0.26 Interpretation

The analysis shows significant differences between weekend and weekday coach prices. Correlation analysis indicates that coach price has a strong positive relationship with miles and a moderate positive relationship with first-class price.

The linear regression model explains about 56.5% of the variation in coach prices, showing that miles, delay, passengers, and hours influence ticket prices. The logistic regression model also indicates that hours and delay significantly affect the probability of redeye flights.

END